

C.2 Extrapolation at $\text{Re}_H = 100$

The models trained at $\text{Re}_H \leq 50$ are evaluated without further training or recalibration at $\text{Re}_H = 100$ for

$$\phi \in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.442649\}.$$

The following figures compare the reference DFT coefficients with the ML predictions for the retained independent $\hat{W}_{m,n}$ and $\hat{V}_{m,n}$ modes.

For $\phi = 0.10$, the reference flow is weakly unsteady and may exhibit symmetry breaking. The comparison at this porosity also shows the limitation of extrapolating a symmetry-constrained model trained on steady flows.

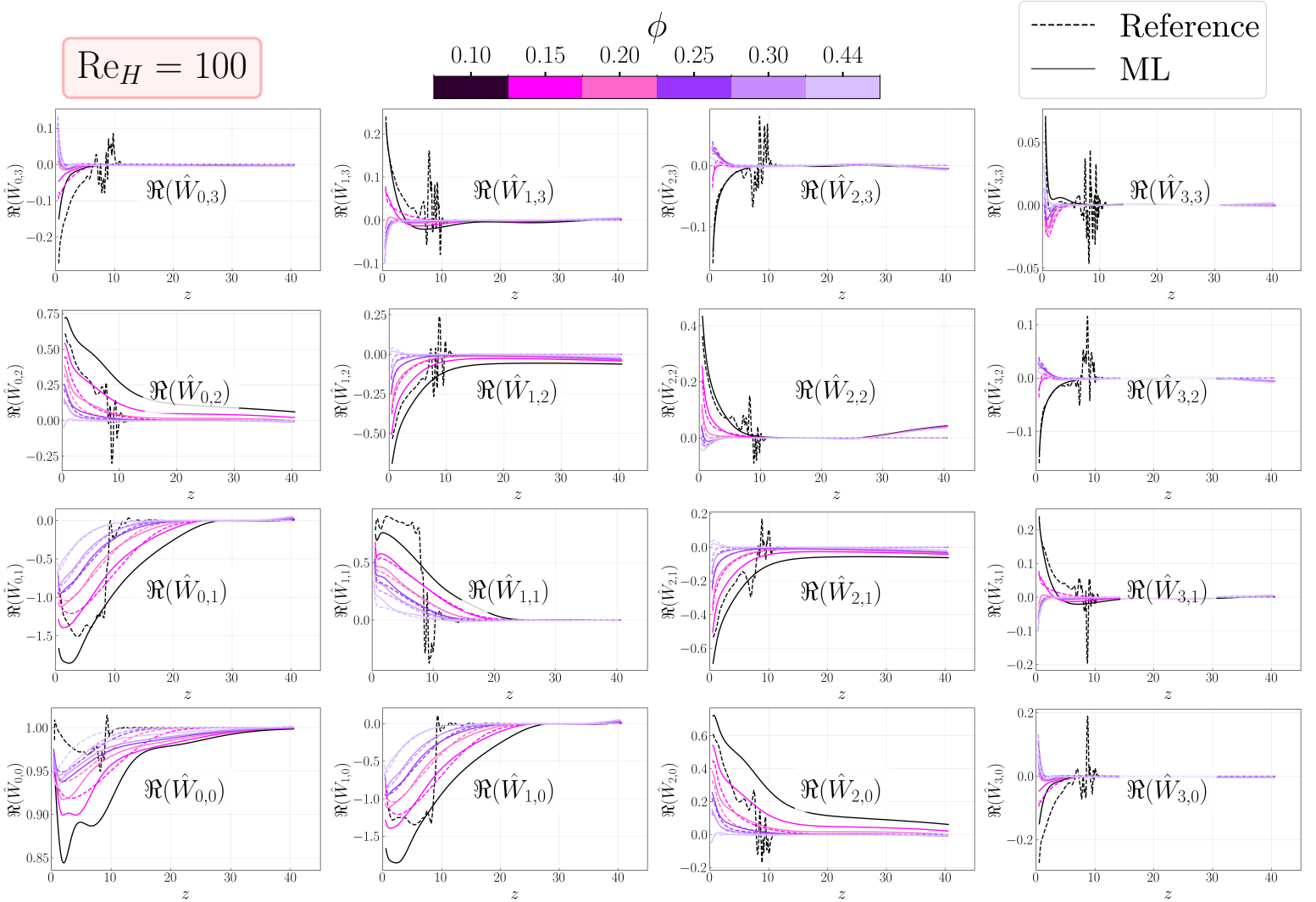


Figure C.44: Comparison of the reference DFT coefficients $\Re(\hat{W}_{m,n})$ (dashed lines) and their neural-network predictions (solid lines) downstream of the SRG at $\text{Re}_H = 100$. Each panel shows a Fourier mode as a function of z ; colors denote the porosity ϕ . This Reynolds number lies outside the training range $\text{Re}_H \leq 50$ and tests extrapolation.

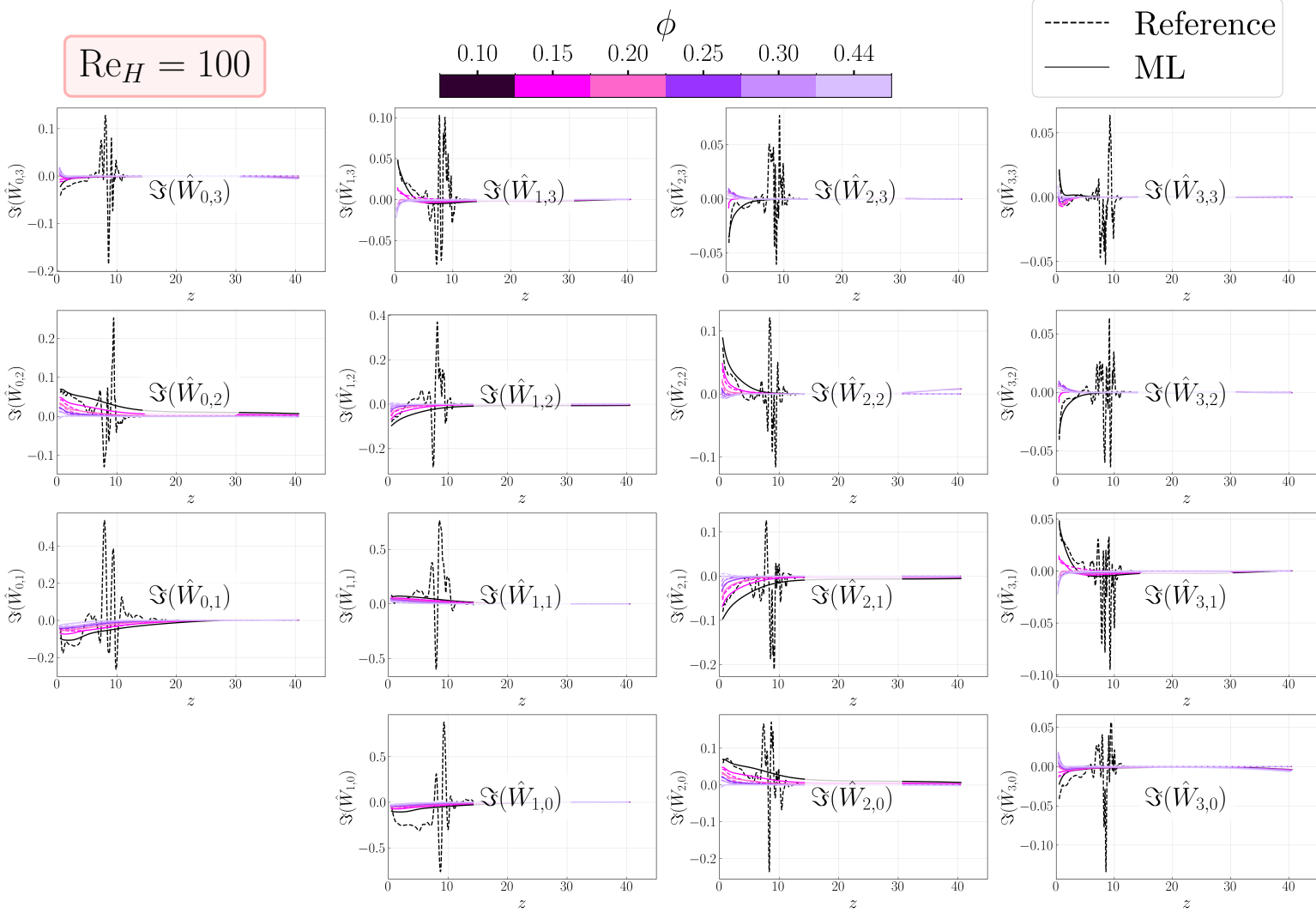


Figure C.45: Comparison of the reference DFT coefficients $\Im(\hat{W}_{m,n})$ (dashed lines) and their neural-network predictions (solid lines) downstream of the SRG at $\text{Re}_H = 100$. Each panel shows a Fourier mode as a function of z ; colors denote the porosity ϕ . This Reynolds number lies outside the training range $\text{Re}_H \leq 50$ and tests extrapolation.

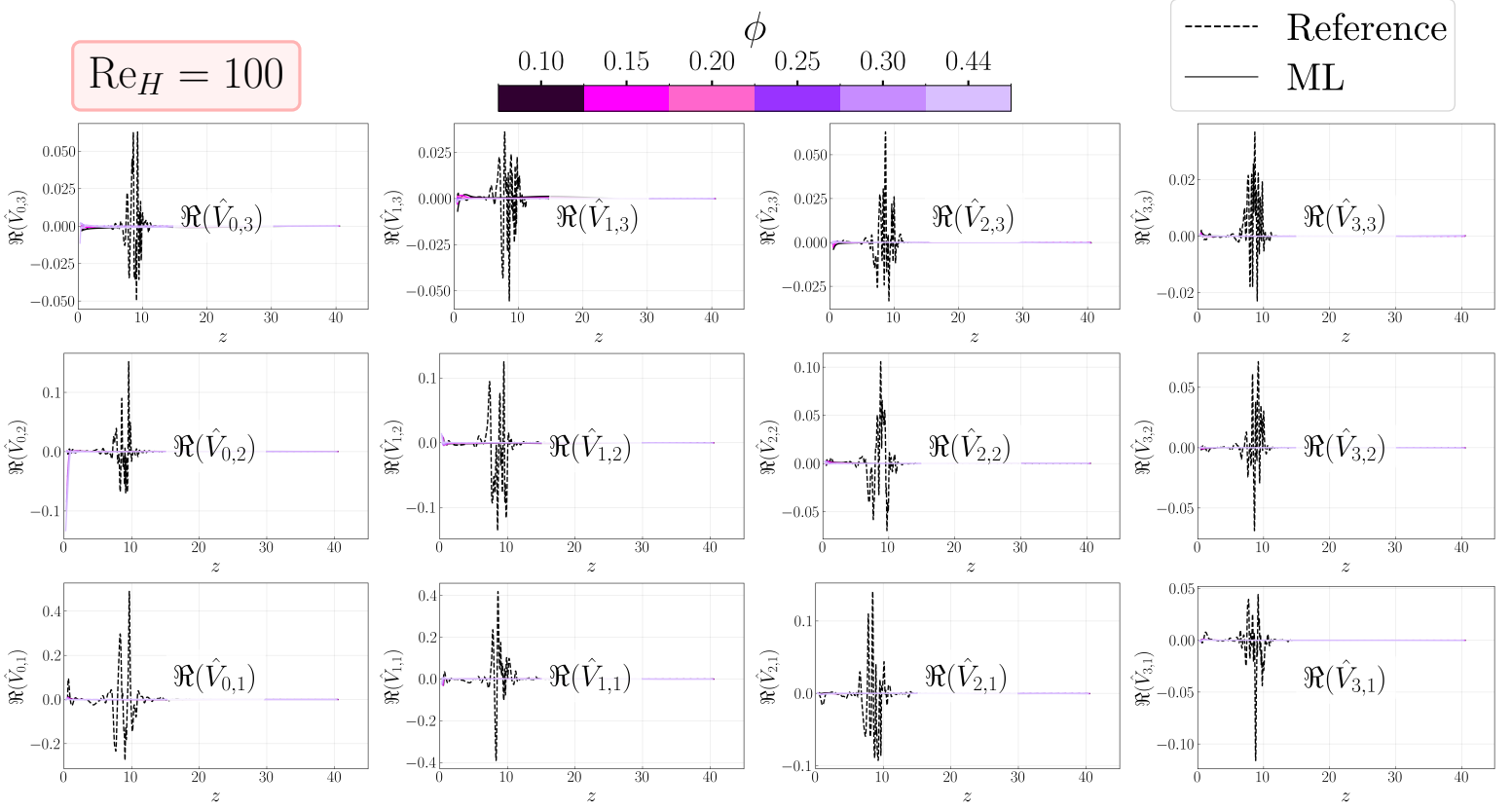


Figure C.46: Comparison of the reference DFT coefficients $\Re(\hat{V}_{m,n})$ (dashed lines) and their neural-network predictions (solid lines) downstream of the SRG at $\text{Re}_H = 100$. Each panel shows a Fourier mode as a function of z ; colors denote the porosity ϕ . This Reynolds number lies outside the training range $\text{Re}_H \leq 50$ and tests extrapolation.

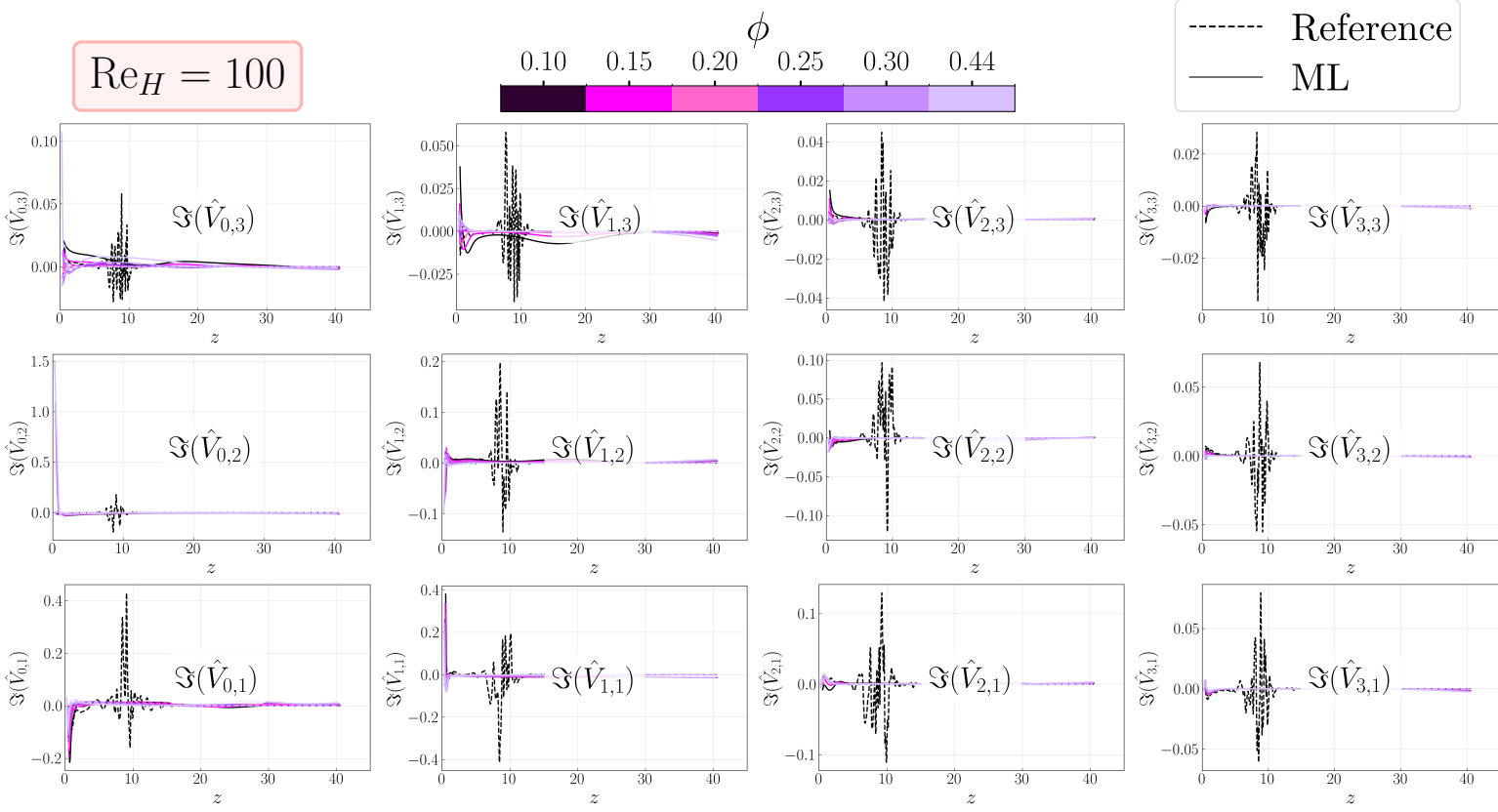


Figure C.47: Comparison of the reference DFT coefficients $\Im(\hat{V}_{m,n})$ (dashed lines) and their neural-network predictions (solid lines) downstream of the SRG at $\text{Re}_H = 100$. Each panel shows a Fourier mode as a function of z ; colors denote the porosity ϕ . This Reynolds number lies outside the training range $\text{Re}_H \leq 50$ and tests extrapolation.

C.3 Training Histories

The following figures show the training and validation losses for the mode-wise regressors. They document convergence and allow comparison of the optimization behavior across the retained Fourier coefficients. The validation curves correspond to samples withheld from optimization within the training parameter range.

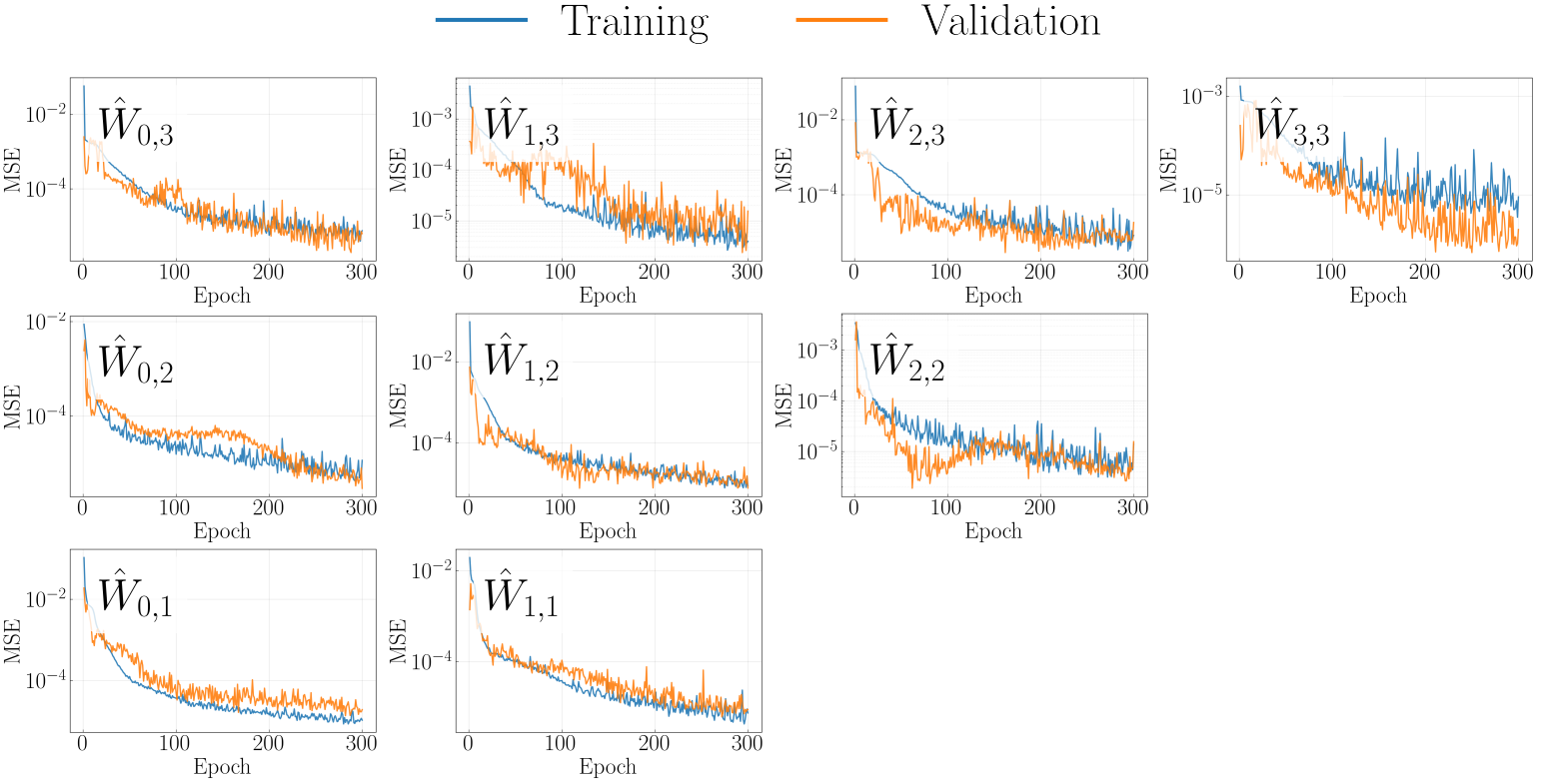


Figure C.48: Training and validation mean-squared-error (MSE) histories for the neural networks predicting the complex Fourier coefficients $\hat{W}_{m,n}$. Each panel represents one retained mode; redundant modes are omitted using the symmetry $\hat{W}_{m,n} = \hat{W}_{n,m}$. The networks use (z, Re_H, ϕ) as inputs and predict both $\Re(\hat{W}_{m,n})$ and $\Im(\hat{W}_{m,n})$. Blue and orange curves show the training and validation MSE, respectively, versus epoch, with MSE on a logarithmic scale. Their decrease and close agreement indicate convergence without clear evidence of overfitting.

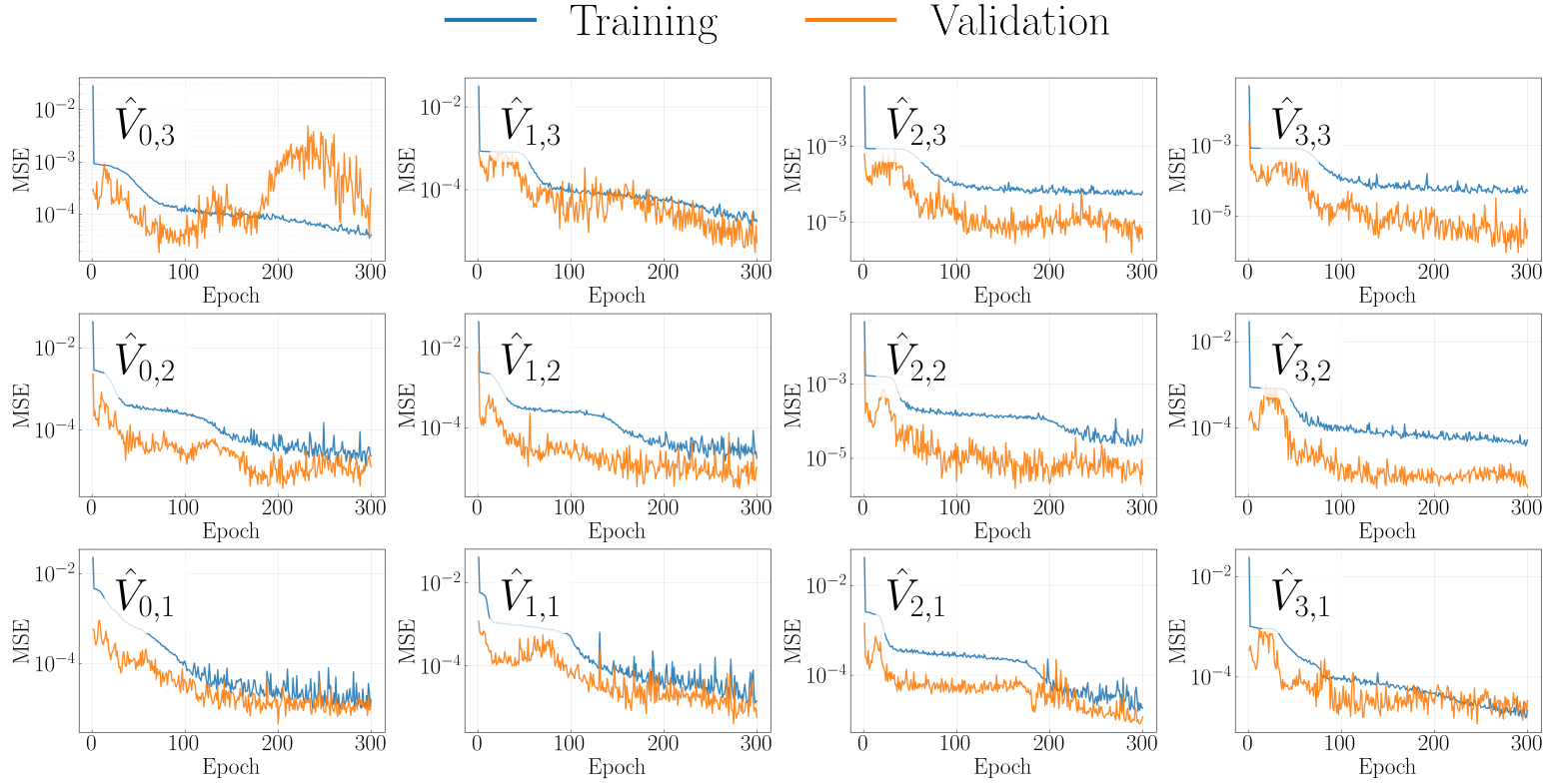


Figure C.49: Training and validation mean-squared-error (MSE) histories for the neural networks predicting the complex Fourier coefficients $\hat{V}_{m,n}$. Each panel represents one retained mode with $m \in \{0, 1, 2, 3\}$ and $n \in \{1, 2, 3\}$. The networks use (z, Re_H, ϕ) as inputs and predict both $\Re(\hat{V}_{m,n})$ and $\Im(\hat{V}_{m,n})$. Blue and orange curves show the training and validation MSE, respectively, versus epoch, with MSE on a logarithmic scale. Their decrease and close agreement indicate convergence without clear evidence of overfitting.

Appendix D

Detailed Validation of the VCDFM Against the SSF

This appendix provides a detailed case-by-case comparison between the geometry-resolved small-scale flow (SSF) and the velocity-corrected Darcy–Forchheimer model (VCDFM) across the investigated porosity and Reynolds-number ranges. For each configuration, the comparison includes: (i) the global streamwise evolution based on cross-sectional L_2 -norms and centerline quantities, (ii) the downstream flow organization in the symmetric (y, z) -plane, (iii) transverse velocity profiles at selected downstream locations, and (iv) transverse (x, y) -plane comparisons.

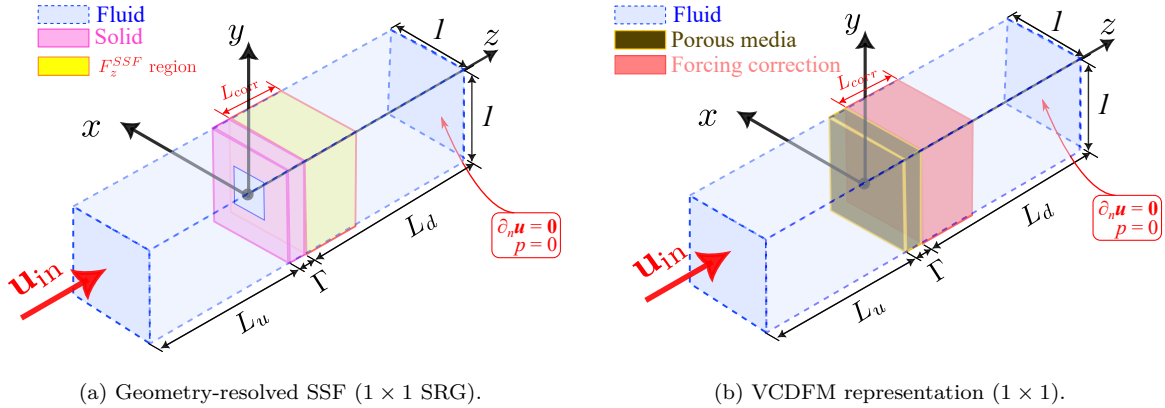


Figure D.1: Comparison between the geometry-resolved small-scale flow (SSF, left) and the velocity-corrected Darcy–Forchheimer model (VCDFM, right). In the VCDFM, a porous slab replaces the grid, and a correction is applied in the downstream region. Lengths are normalized by the grid spacing H ; the transverse unit cell therefore has size 1×1 .

In the VCDFM, the grid is represented by a porous slab, and a streamwise correction is applied downstream to recover the velocity redistribution observed in the geometry-resolved SSF. The correction is constructed from the SSF velocity field $\mathbf{u} = (u, v, w)$ through

$$F_z = -(\mathbf{u} \cdot \nabla)w - \frac{1}{\text{Re}_H} \nabla^2 w. \quad (\text{D.1})$$

Here, w is the streamwise velocity and Re_H is the Reynolds number based on the grid

spacing H .

The velocity-correction source acts only in the streamwise direction:

$$\mathbf{S}^{(\text{VC})} = (0, 0, -\zeta \Lambda^{(\text{VC})} F_z), \quad (\text{D.2})$$

where ζ controls the correction intensity and

$$\Lambda^{(\text{VC})} = \begin{cases} 1, & \Gamma/2 \leq z \leq \Gamma/2 + L_{\text{corr}}, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{D.3})$$

The dimensionless grid thickness is $\Gamma = t/H$, and L_{corr} is the downstream extent of the correction. The SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM variants use forcing reconstructed from the geometry-resolved SSF, truncated Fourier modes, and machine-learning predictions, respectively.

D.1 Effective Validity Range of the VCDFM

This section summarizes the effective regime of validity of the velocity-corrected Darcy–Forchheimer model (VCDFM) across the investigated porosity and Reynolds-number ranges. For each porosity, comparisons between the geometry-resolved small-scale flow (SSF) and the VCDFM are reported in the symmetric (y, z) -plane over a range of Reynolds numbers. The highlighted regions indicate the Reynolds-number range for which noticeable deviations between the VCDFM and SSF emerge, marking the onset of model breakdown and loss of predictive accuracy.

For the SRG configuration, the predictive accuracy of the VCDFM depends strongly on both porosity and forcing reconstruction strategy. For $\phi = 0.1$, noticeable deviations between the geometry-resolved small-scale flow (SSF) and VCDFM emerge at $\text{Re}_H = 33.33$, 50, and 100 for the SSF-VCDFM and TIDFT-VCDFM reconstructions (Figs. D.3 and D.4), whereas ML-VCDFM exhibits an earlier loss of predictive accuracy at $\text{Re}_H = 33.33$ and 50 (Fig. D.5).

As the porosity increases to $\phi = 0.15$, deviations shift toward higher Reynolds numbers, appearing only at $\text{Re}_H = 50$ and 100 for the SSF-VCDFM and TIDFT-VCDFM reconstructions (Figs. D.6 and D.7), while ML-VCDFM is evaluated up to $\text{Re}_H = 50$, for which noticeable deviations become visible (Fig. D.8).

For $\phi = 0.2$, deviations become noticeable only at $\text{Re}_H = 100$ for both SSF-VCDFM and TIDFT-VCDFM (Figs. D.9 and D.10), whereas no significant departure is observed for ML-VCDFM within the investigated range up to $\text{Re}_H = 50$.

Finally, for $\phi = 0.25$, no significant departure between VCDFM and SSF is observed for the SSF-VCDFM and TIDFT-VCDFM variants across the considered Reynolds-number range (Figs. D.12 and D.13). ML-VCDFM also remains accurate up to $\text{Re}_H = 50$, but shows significant deviations at the extrapolation case $\text{Re}_H = 100$ (Fig. D.14).

Overall, the results indicate that the predictive capability of the VCDFM improves with increasing porosity, whereas lower-porosity configurations experience an earlier deterioration of model accuracy as inertial effects strengthen and transverse velocity redistribution becomes increasingly pronounced.

The effective validity regime identified throughout the chapter is summarized in Fig. D.2. The map classifies each investigated (ϕ, Re_H) configuration according to

whether the VCDFM remains capable of recovering the dominant downstream velocity redistribution and transverse flow organization observed in the geometry-resolved small-scale flow (SSF).

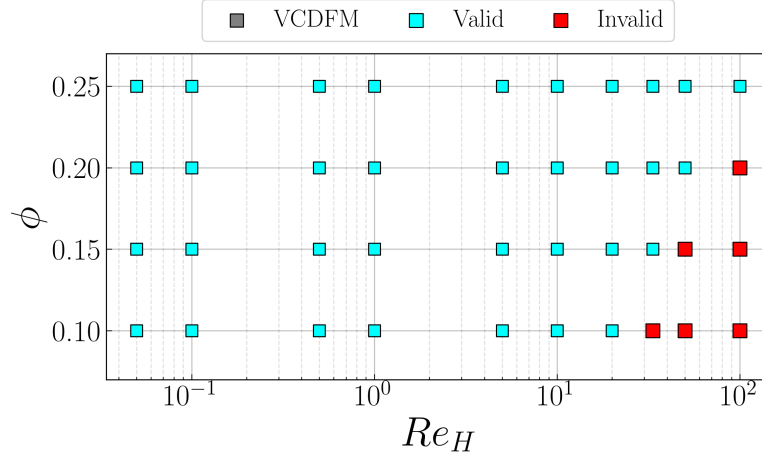


Figure D.2: Validity map of the VCDFM for the 1×1 SRG across the investigated Reynolds numbers and porosities. Cyan markers indicate cases within the effective validity regime; red markers indicate cases where the VCDFM correction is no longer physically consistent.

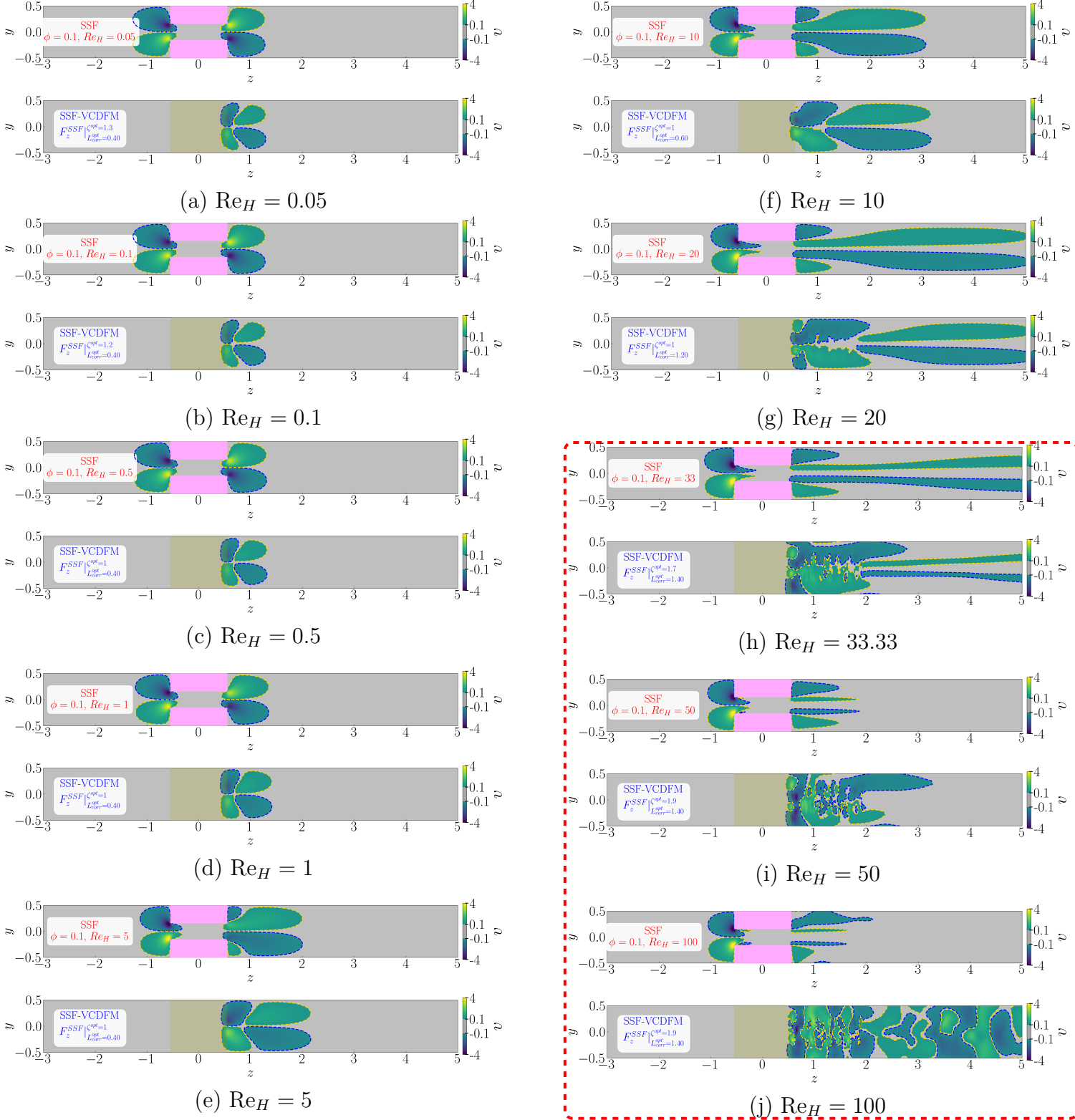
$\phi = 0.10$


Figure D.3: Transverse velocity component v in the streamwise (z, y) -plane for the SRG configuration at $\phi = 0.10$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the SSF-VCDFM reconstruction using F_z^{SSF} . The dashed red rectangle highlights $Re_H = 33.33, 50$, and 100 , where noticeable differences between the SSF-VCDFM and SSF indicate a loss of predictive accuracy at $\phi = 0.10$.